Supporting Information

Detailed methods for "Income inequality increases optimal carbon price"

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This Supporting Information specifies the model, the numerical procedures, the design of the simulation experiments, the estimation of the model's exogenous inputs and the construction of the figures. Together with the configuration file of each run and the results archived with the code, it is intended to determine every reported number. A companion document in the code repository (`methods/METHODS_DETAIL.md`) gives the same specification at the level of the implementation, including the floating-point evaluation order needed to reproduce results bit for bit.

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S1 Overview and reproducibility

The model is a stylized, globally aggregated climate–economy model in the tradition of DICE [1] and COIN [2], extended with a continuous income distribution. Output is produced by a Cobb–Douglas technology and is reduced by climate damage and by the cost of emission abatement. Damage is distributed across the income distribution according to a vulnerability exponent β, and the cost of abatement is raised by a tax whose progressivity is set by a parameter τ. Warming is computed from cumulative emissions with a three-mode impulse-response model. Two control trajectories, the carbon price (equivalently the marginal abatement cost) and the savings rate, are chosen to maximize the discounted sum of population-weighted mean utility.

The calculation has three stages, which are reproducible to different degrees.

1. *Forward integration* of the model for given control trajectories (Sections S3–S7) is deterministic. Replaying the published optimal controls through the model reproduces every published time series to a relative difference of 3.4×10^{-13} or less.

2. *The objective* (Section S8) inherits the precision of the forward integration.

3. *The optimization* (Sections S9 and S10) uses a deterministic derivative-free algorithm, but the path it takes depends on every bit of every objective value, so implementations that differ only by round-off end at slightly different points. Re-optimizing manuscript runs with code whose objective values differ from the original only by round-off reproduced the objective to 10^{-13} (relative), the control f to about 3×10^{-6}, and the carbon price in 2030 and 2100 and the warming in 2100 to 2×10^{-6} (relative).

An independent reimplementation of the model and the optimizer, written only from the repository specification without access to the model's source code, reproduced the regression baseline (Section S14) bit for bit.

All arithmetic is IEEE-754 double precision. The reference results were produced with Python 3.10.12, NumPy 2.2.6 [4], SciPy 1.15.3 [5] and the NLopt optimization library [6].

S2 Conventions and notation

Time. The model time t is the calendar year. Exogenous time functions and control times are expressed in relative time $t_r = t - t_\mathrm{b}$, with $t_\mathrm{b} = 2020$. The integration runs from $t_0 = 2020$ to $t_\mathrm{end} = 2420$ with time step $\Delta t = 1$ yr.

Units. Monetary quantities are in 2019 US dollars (the DICE-2023 convention), and income is in dollars per person per year. Emissions are in $\mathrm{tCO_2}$ (cumulative) or $\mathrm{tCO_2\ yr^{-1}}$ (rates), and temperature is in °C above preindustrial.

Per-capita and aggregate quantities. Lower-case symbols denote per-capita quantities and upper-case symbols denote totals; for example, gross output is $Y_\mathrm{g} = y_\mathrm{g} L$ for population L.

Population rank. $F \in [0, 1]$ is the rank in the income distribution, with $F = 0$ the poorest. The Lorenz curve $\mathcal{L}(F)$ is the share of total income received by the poorest fraction F of the population, so that income at rank F is proportional to $\mathcal{L}'(F)$, which has mean one.

Clipping. We write $\mathrm{clip}(x; a, b) = \min\{\max\{x, a\}, b\}$. Several quantities are bounded away from zero or one by the small constant $\varepsilon = 10^{-12}$; these safeguards are collected in Section S15.

Principal symbols. Table S1 lists the symbols used in the model equations.

Table S1: Principal symbols.

Symbol	Meaning
K	capital stock (\$)
A, L	total factor productivity; population (persons)
$y_\mathrm{g}, y_\mathrm{d}, y_\mathrm{n}$	per-capita gross output, output after climate damage, and output after damage and abatement cost (\$ person^{-1} yr^{-1})
y_a	per-capita abatement expenditure
Ω_b	aggregate damage fraction implied by the current warming
Ω, ω_i	aggregate and per-node damage fractions

Symbol	Meaning
q_i, Q	per-node and aggregate damage ratios carried to the next step
f, P	control $f = \log_{10} P$; carbon price P (\$ $\mathrm{tCO_2^{-1}}$), equal to the marginal abatement cost
s	savings rate (control)
μ	abatement fraction
$\sigma, \xi, E_{\mathrm{land}}$	CO_2 intensity of output; ratio of CO_2-equivalent to industrial CO_2 emissions; land-use emissions
θ_1, θ_2	backstop price (\$ $\mathrm{tCO_2^{-1}}$); abatement cost exponent
E_{cum}, H_k	cumulative emissions; temperature contributions of the three impulse-response modes
ΔT	warming above preindustrial
G	Gini coefficient of pre-damage income
$\mathcal{L}, \mathcal{L}_b$	Lorenz curve; base (empirical) Lorenz curve
F_i, w_i	quadrature nodes and weights in population rank, $i = 1, \ldots, N_q$
η, ρ	elasticity of marginal utility; pure rate of time preference
τ, η_τ	tax progressivity parameter; exponent of the tax schedule
β	damage vulnerability exponent
ΔU	utility loss parameter of the tax schedule
U	population-weighted mean utility, $U = L\,\bar{u}$
ε	numerical safeguard constant, 10^{-12}

S3 Exogenous inputs

S3.1 Scalar parameters

Each run is specified by a configuration that fixes all parameters; there are no implicit defaults. Table S2 lists the scalar parameters. Only η, ρ, τ and β vary between the manuscript runs (Section S11).

Table S2: Scalar parameters of the manuscript runs.

Symbol	Value	Meaning
α	0.3	capital share of output
δ	0.1 $\mathrm{yr^{-1}}$	capital depreciation rate
ψ_1	0 $\mathrm{^\circ C^{-1}}$	linear damage coefficient
ψ_2	0.003467 $\mathrm{^\circ C^{-2}}$	quadratic damage coefficient
$\Omega_{\max}$	0.8	upper bound on the damage fraction
η	1.5 (varied)	elasticity of marginal utility
ρ	0.025 $\mathrm{yr^{-1}}$ (varied)	pure rate of time preference
θ_2	2.6	abatement cost exponent
$\mu_{\max}$	10^{20}	upper bound on μ (effectively none)
K_0	2.95×10^{14} \$	capital in 2020
y_{ref}	17 210.79 \$ $\mathrm{person^{-1}\ yr^{-1}}$	reference income of the damage distribution
β	0, 0.5, 1 (varied)	damage vulnerability exponent
τ	0, 0.5, 0.998 (varied)	tax progressivity, $0 \le \tau < 1$
N_q	32	number of quadrature nodes

$\mu > 1$ is allowed and represents net-negative industrial emissions. The model also supports emission and consumption pulses at a given year (Section S5.5); their amplitudes are zero in all manuscript runs.

S3.2 Time functions

Exogenous time-dependent inputs are given by one of the following functional forms, evaluated at $t_r = t - t_\mathrm{b}$:

$$\text{exponential:} \quad g(t_r) = a\, e^{b t_r} + c, \tag{S1}$$

$$\text{double exponential:} \quad g(t_r) = g_0 \left[\phi\, e^{b_1 t_r} + (1-\phi)\, e^{b_2 t_r}\right], \tag{S2}$$

$$\text{Gompertz:} \quad g(t_r) = g_\infty \exp\left[\ln\left(\frac{g_0}{g_\infty}\right) e^{c\, t_r}\right], \tag{S3}$$

$$\text{logistic:} \quad g(t_r) = \frac{g_\infty}{1 + (g_\infty/g_0 - 1)\, e^{-b t_r}}, \tag{S4}$$

$$\text{piecewise linear:} \quad \text{linear interpolation between given points, constant beyond the end points,} \tag{S5}$$

or a constant. The Gompertz form has $g(0) = g_0$ and $g \to g_\infty$ for $c < 0$. Table S3 gives the time functions of the manuscript runs; Section S12 gives their provenance.

Table S3: Exogenous time functions of the manuscript runs ($t_r = t - 2020$).

Input	Form	Parameters
$A(t)$	Gompertz	$g_0 = 739.619$, $g_\infty = 4\,906\,815.811$, $c = -0.0015$ yr^{-1}
$L(t)$ (persons)	Gompertz	$g_0 = 7.752 \times 10^9$, $g_\infty = 1.0825 \times 10^{10}$, $c = -0.03133$ yr^{-1}
$\sigma(t)$ (tCO$_2$ \$$^{-1}$)	Gompertz	$g_0 = 2.91355 \times 10^{-4}$, $g_\infty = 4.4681 \times 10^{-5}$, $c = -0.008164$ yr^{-1}
$\theta_1(t)$ (\$ tCO$_2^{-1}$)	double exponential	$g_0 = 6035.6$, $\phi = 0.906$, $b_1 = -0.224$, $b_2 = -0.00112$ yr^{-1}
$G(t)$	exponential	$a = 0.19517$, $b = -0.01058$ yr^{-1}, $c = 0.37796$
$\xi(t)$	exponential	$a = 1.4$, $b = -0.002085$ yr^{-1}, $c = 0$
$E_\mathrm{land}(t)$ (tCO$_2$ yr^{-1})	exponential	$a = 5.9 \times 10^9$, $b = -0.02107$ yr^{-1}, $c = 0$

The Gini trajectory gives $G(2020) = 0.57313$ and $G(\infty) = 0.37796$. In the runs that compare the model with DICE-2023 (Section S11), $G(t) \equiv 0$.

S3.3 Climate response and initial state

Warming responds to cumulative CO_2 emissions through the impulse response of the median model of Ricke and Caldeira [3]. The warming at time t caused by a unit emission at time t' is

$$R(t - t') = R_\infty + \sum_{k=1}^{3} a_k\, e^{-(t-t')/\zeta_k}, \qquad R_\infty = -(a_1 + a_2 + a_3), \tag{S6}$$

with $a = (-0.629,\ 0.203,\ -0.0521) \times 10^{-12}$ °C tCO$_2^{-1}$ and $\zeta = (2.241,\ 35.750,\ 97.180)$ yr, so that $R_\infty = 4.781 \times 10^{-13}$ °C tCO$_2^{-1}$ and $R(0) = 0$.

The 2020 climate state comes from a fit of historical warming,

$$\Delta T_{\mathrm{hist}}(t) = 1.1126 - 0.07022\, e^{(2020-t)/\zeta_1} + 0.21036\, e^{(2020-t)/\zeta_2} - 0.08308\, e^{(2020-t)/\zeta_3} \quad (^\circ\mathrm{C}), \quad \text{(S7)}$$

obtained by convolving an exponential emission history (633.5 GtC cumulative to 2020, as in DICE-2023) with Eq. (S6). Its constant term is $R_\infty E_{\mathrm{cum}}(2020)$, which gives $E_{\mathrm{cum}}(2020) = 1.1126/R_\infty = 2.327128 \times 10^{12}$ $\mathrm{tCO_2}$. The other three terms are the 2020 mode temperatures, $H_1 = -0.07022$, $H_2 = 0.21036$ and $H_3 = -0.08308$ °C. The initial warming is $\Delta T(2020) = 1.16966$ °C.

S4 Income distribution

S4.1 Quadrature over population rank

Population averages are computed with N_q-point Gauss–Legendre quadrature mapped to $[0, 1]$. If x_i and W_i are the Gauss–Legendre nodes and weights on $[-1, 1]$, the rank nodes and population weights are

$$F_i = \tfrac{1}{2}(x_i + 1), \qquad w_i = \tfrac{1}{2} W_i, \qquad \sum_{i=1}^{N_q} w_i = 1, \quad \text{(S8)}$$

and the population mean of a quantity z is $\langle z \rangle = \sum_i w_i z_i$. Node i represents the population interval $[F_{i-}, F_{i+}]$ with $F_{1-} = 0$ and $F_{i+} - F_{i-} = w_i$; by the Chebyshev–Markov–Stieltjes separation theorem each node lies strictly inside its interval, so node values also serve as piecewise-constant values on the intervals. For $N_q = 32$, $F_1 = 0.001368$, $F_{32} = 0.998632$ and $w_1 = 0.003509$.

S4.2 Empirical Lorenz curve

All manuscript runs use a Lorenz curve fitted to the global income distribution (Section S12.2). The base curve and its derivative are

$$\mathcal{L}_b(F) = \sum_{k=0}^{3} \gamma_k F^{p_k}, \qquad \mathcal{L}_b'(F) = \sum_{k=0}^{3} \gamma_k\, p_k\, F^{p_k - 1}, \quad \text{(S9)}$$

with exponents and weights

$$p = (1.500036,\ 4.367440,\ 14.072005,\ 135.059674),$$
$$\gamma_1 = 0.3776187268483524, \quad \gamma_2 = 0.3671247620949191,$$
$$\gamma_3 = 0.09538538350961864, \quad \gamma_0 = 1 - \gamma_1 - \gamma_2 - \gamma_3.$$

Its Gini coefficient is

$$G_b = 1 - 2 \sum_{k=0}^{3} \frac{\gamma_k}{p_k + 1} = 0.681279513926459. \quad \text{(S10)}$$

A Lorenz curve with any Gini coefficient $0 \le G \le G_b$ is obtained by mixing the base curve linearly with perfect equality:

$$\mathcal{L}(F) = (1 - a)\, F + a\, \mathcal{L}_b(F), \qquad \mathcal{L}'(F) = (1 - a) + a\, \mathcal{L}_b'(F), \qquad a = G/G_b, \quad \text{(S11)}$$

which has Gini coefficient exactly G. A Gini coefficient above G_b is treated as a configuration error. For $G = 0$, $\mathcal{L}'(F) \equiv 1$.

The density $\mathcal{L}_b'$ is steep at the top of the distribution ($p_3 = 135$), so the quadrature does not integrate it exactly: the computed mean $\sum_i w_i \mathcal{L}'(F_i) - 1$ is 3.2×10^{-7} at $G = 0.3$, 6.05×10^{-7} at $G = 0.573$ and 7.2×10^{-7} at $G = G_b$. The model does not renormalize; this error is part of the published results.

The model also supports a Pareto Lorenz curve, $\mathcal{L}'(F) = (1 - 1/a)(1 - F)^{-1/a}$ with $a = (1 + 1/G)/2$. It is not used for any result: its density is singular at $F \to 1$, and with 32 nodes the quadrature loses 2–10% of income.

S5 Climate and economy in one time step

At each time t the model computes the tendencies of the state variables $(K, E_{\text{cum}}, H_1, H_2, H_3)$ from the state, the exogenous inputs, the controls $f(t)$ and $s(t)$, and two damage-distribution ratios carried from the previous step: the per-node ratios q_i and the aggregate ratio Q. The calculation proceeds in the order of the following subsections.

S5.1 Temperature

$$\Delta T = R_\infty E_{\text{cum}} + H_1 + H_2 + H_3. \tag{S12}$$

S5.2 Climate damage with a lagged distribution

The aggregate damage fraction implied by the current warming is

$$\Omega_{\text{b}} = \min\left\{\Omega_{\max},\ \psi_1 \Delta T + \psi_2 \Delta T^2\right\}. \tag{S13}$$

The damage applied in this step uses the current Ω_{b} but the distribution of damage across incomes computed at the end of the previous step (Section S5.9):

$$\tilde{\Omega} = \text{clip}(Q\,\Omega_{\text{b}};\ 0,\ 1-\varepsilon), \qquad \tilde{\omega}_i = \text{clip}(q_i\,\Omega_{\text{b}};\ 0,\ 1-\varepsilon). \tag{S14}$$

In the first step $q_i = Q = 1$, i.e. damage is uniform across incomes.

S5.3 Production

$$y_{\text{g}} = A\left(\frac{K}{L}\right)^{\alpha}, \qquad y_{\text{d}} = (1 - \tilde{\Omega})\, y_{\text{g}}, \tag{S15}$$

with $y_{\text{g}} = 0$ if $K \le 0$. Climate damage per capita is $\tilde{\Omega}\, y_{\text{g}}$. If $y_{\text{g}} \le \varepsilon$ or $y_{\text{d}} \le \varepsilon$ the economy has collapsed and the step is handled as described in Section S5.11.

S5.4 Abatement

The control sets the carbon price, $P = 10^f$, which equals the marginal abatement cost. Potential (unabated) emissions per capita, including land-use emissions, are

$$e_{\text{pot}} = \sigma\, y_{\text{g}}\, \xi + E_{\text{land}}/L. \tag{S16}$$

Abatement follows the DICE cost curve, whose marginal cost $\theta_1 \mu^{\theta_2 - 1}$ is set equal to P:

$$\mu = \min\left\{\mu_{\max},\ \left(\frac{P}{\theta_1}\right)^{1/(\theta_2 - 1)}\right\}, \qquad y_{\text{a}} = \min\left\{\frac{\theta_1}{\theta_2}\, e_{\text{pot}}\, \mu^{\theta_2},\ (1-\varepsilon)\, y_{\text{d}}\right\}, \tag{S17}$$

if $e_{\text{pot}} > \varepsilon$ and $P > \varepsilon$; otherwise $\mu = y_{\text{a}} = 0$. Abatement expenditure is paid from damaged output:

$$\lambda = y_{\text{a}}/y_{\text{d}}, \qquad y_{\text{n}} = (1-\lambda)\, y_{\text{d}}. \tag{S18}$$

Land-use emissions are abated at the same fraction μ as industrial emissions.

S5.5 Emission and consumption pulses

For impulse experiments, an emission pulse X_E (tCO_2) or a consumption pulse X_C (\$) can be added in the step $[t, t+\Delta t)$ that contains a given year. They contribute $E_{\text{extra}} = X_E/\Delta t$ to the emission rate and $c_{\text{extra}} = X_C/(L\Delta t)$ to per-capita income in that step, and zero otherwise. Both are zero in all manuscript runs.

S5.6 Pre-tax income distribution

Pre-tax income at node i is gross income at that rank, reduced by the node's lagged damage fraction:

$$y_{\text{pre},i} = y_{\text{g}}\, \mathcal{L}'(F_i)\, (1-\tilde{\omega}_i), \qquad \hat{y}_i = y_{\text{pre},i}/y_{\text{d}}, \tag{S19}$$

where $\mathcal{L}'$ uses the current $G(t)$, and $\hat{y}_i$ is pre-tax income normalized by mean damaged output. The mean $\langle \hat{y} \rangle$ equals one only approximately, because of the quadrature error of Section S4.2 and because the lagged node damages $\tilde{\omega}_i$ were normalized with the previous step's post-tax incomes (Section S5.9), whereas $\tilde{\Omega}$ is applied in aggregate. The model uses these quantities as they are.

S5.7 Progressive taxation

The abatement expenditure is financed by a tax on post-damage income. The per capita revenue required is $T = y_{\text{a}} - c_{\text{extra}}$ (so a consumption pulse reduces the tax), or $\hat{T} = T/y_{\text{d}}$ in normalized units.

Tax schedule. The tax belongs to the family of equal-utility-loss schedules: every person loses the same amount $\Delta U \geq 0$ of utility measured with an isoelastic function of elasticity η_τ, $u_\tau(c) = c^{1-\eta_\tau}/(1-\eta_\tau)$ (or $\ln c$ for $\eta_\tau = 1$). The progressivity parameter $\tau \in [0, 1)$ sets

$$\eta_\tau = 1 + \frac{\tau}{1-\tau}\,(\eta - 1). \tag{S20}$$

$\tau = 0$ gives $\eta_\tau = 1$, a flat tax rate; $\tau = 0.5$ gives $\eta_\tau = \eta$, equal utility loss under the welfare function; and $\tau \to 1$ gives $\eta_\tau \to \infty$, a tax that falls on the highest incomes. For $\eta = 1.5$ and $\tau = 0.998$, $\eta_\tau = 250.5$.

Post-tax normalized income $c(\hat{y}; \Delta U)$ solves $u_\tau(\hat{y}) - u_\tau(c) = \Delta U$:

$$c(\hat{y}; \Delta U) = \begin{cases} \hat{y}\, e^{-\Delta U}, & \eta_\tau = 1, \\ \left[\hat{y}^{\,1-\eta_\tau} + (\eta_\tau - 1)\, \Delta U\right]^{-1/(\eta_\tau - 1)}, & \eta_\tau > 1, \\ \left[\max\left\{\hat{y}^{\,1-\eta_\tau} - (1-\eta_\tau)\, \Delta U,\ 0\right\}\right]^{1/(1-\eta_\tau)}, & \eta_\tau < 1. \end{cases} \tag{S21}$$

The case $\eta_\tau = 1$ is applied when $|\eta_\tau - 1| < \varepsilon$. For $\eta_\tau > 1$ the schedule is evaluated in logarithms,

$$\ln c = \ln \hat{y} - \frac{1}{\eta_\tau - 1}\, \ln[1 + \exp(\ln(\eta_\tau - 1) + \ln \Delta U + (\eta_\tau - 1) \ln \hat{y})]\,, \tag{S22}$$

with $\ln \Delta U$ as the unknown. For strongly progressive taxes ΔU is extremely small (for $\tau = 0.998$, $\ln \Delta U$ lies between about -670 and -460), and working with $\ln \Delta U$ avoids an underflow that would otherwise switch the tax off.

Revenue condition. If $T \le \varepsilon$ there is no tax. For a flat tax ($|\eta_\tau - 1| < \varepsilon$), every income is reduced by the fraction $r = \min\{\hat{T}, 1-\varepsilon\}$, i.e. $\Delta U = -\ln(1-r)$. This closed form assumes $\langle \hat{y} \rangle = 1$, so the revenue raised differs from $\hat{T}$ by the small relative error discussed above. Otherwise $x = \ln \Delta U$ is the root of the revenue residual

$$g(x) = \sum_i w_i \left[\hat{y}_i - c\left(\hat{y}_i; e^x\right) \right] - \hat{T}, \tag{S23}$$

found with Brent's method [8] on the bracket

$$x_{\mathrm{lo}} = -\left| (\eta_\tau - 1) \ln \max_i \hat{y}_i \right| - 50, \qquad x_{\mathrm{hi}} = \ln 10^{10}, \tag{S24}$$

with an absolute tolerance of 10^{-9} in x, a relative tolerance of $4 \cdot 2^{-52}$ and at most 100 iterations. If $g(x_{\mathrm{lo}}) \ge 0$, the target is negligible and no tax is levied. If $g(x_{\mathrm{hi}}) < 0$, the revenue cannot be raised and the control trajectory is infeasible (Section S10.3). A solution whose post-tax incomes are not finite, or decrease with rank by more than 10^{-9}, is also treated as a failure.

Post-tax income is $y_{\mathrm{n},i} = c(\hat{y}_i; \Delta U)\, y_{\mathrm{d}}$.

Tax-rate diagnostics. The average and marginal tax rates at node i are reported but do not affect the dynamics:

$$t_i^{\mathrm{avg}} = 1 - \frac{y_{\mathrm{n},i}}{y_{\mathrm{pre},i}}, \qquad t_i^{\mathrm{marg}} = 1 - \frac{\partial c}{\partial \hat{y}} = 1 - \left(\frac{y_{\mathrm{n},i}}{y_{\mathrm{pre},i}} \right)^{\eta_\tau}, \tag{S25}$$

with the ratio $y_{\mathrm{n},i}/y_{\mathrm{pre},i}$ set to one where $y_{\mathrm{pre},i} \le \varepsilon$, and the marginal rate set to one where that ratio is at most ε.

S5.8 Utility

Every person saves the same fraction s of post-tax income. Consumption and utility at node i are

$$c_i = (1-s)\, y_{\mathrm{n},i}, \qquad u_i = \begin{cases} \dfrac{\max\{c_i, \varepsilon\}^{1-\eta}}{1-\eta}, & \eta \neq 1, \\ \ln \max\{c_i, \varepsilon\}, & \eta = 1, \end{cases} \tag{S26}$$

and the population-weighted mean utility is

$$U = L\, \bar{u}, \qquad \bar{u} = \sum_i w_i u_i. \tag{S27}$$

S5.9 Damage distribution for the next step

Damage fractions are distributed across incomes by a power law in post-tax income,

$$\omega_i^* = \mathrm{clip}\left(\Omega_{\mathrm{b}} \left(\frac{y_{\mathrm{n},i}}{y_{\mathrm{ref}}} \right)^{-\beta};\ 0,\ 1-\varepsilon \right), \tag{S28}$$

with $\omega_i^* = \Omega_{\mathrm{b}}$ for $\beta = 0$ (applied when $|\beta| < \varepsilon$) and $\omega_i^* = 1 - \varepsilon$ where $y_{\mathrm{n},i} \le \varepsilon$. The fractions are then rescaled so that the income-weighted mean damage fraction equals Ω_{b}:

$$\omega_i = \mathrm{clip}\left(\omega_i^* \frac{\Omega_{\mathrm{b}} \sum_j w_j y_{\mathrm{n},j}}{\sum_j w_j\, y_{\mathrm{n},j}\, \omega_j^*};\ 0,\ 1-\varepsilon \right), \qquad \Omega = \frac{\sum_i w_i\, y_{\mathrm{n},i}\, \omega_i}{\sum_i w_i\, y_{\mathrm{n},i}}. \tag{S29}$$

Thus β redistributes damage across incomes without changing its aggregate, and y_{ref} cancels except through the clipping bounds. For $\beta = 1$, absolute damage $\omega_i\, y_{\mathrm{n},i}$ is the same for every person; for $\beta > 0$, damage falls disproportionately on lower incomes. (The rescaling is skipped in an optional variant, not used for any result, in which the aggregate damage depends on the income distribution.)

The ratios carried to the next step are

$$q_i = \omega_i/\Omega_{\mathrm{b}}, \qquad Q = \Omega/\Omega_{\mathrm{b}}, \tag{S30}$$

or zero if $\Omega_{\mathrm{b}} \le \varepsilon$. The next step multiplies them by its own Ω_{b} (Eq. S14): the temperature dependence of damage is current, while its distribution across incomes lags by one time step.

S5.10 Emissions, capital and climate tendencies

Net emissions per capita and the tendencies of the state are

$$e = (1-\mu)\, e_{\mathrm{pot}}, \tag{S31}$$

$$\frac{\mathrm{d}K}{\mathrm{d}t} = s\, y_{\mathrm{n}} L - \delta K, \tag{S32}$$

$$\frac{\mathrm{d}E_{\mathrm{cum}}}{\mathrm{d}t} = e\, L + E_{\mathrm{extra}}, \tag{S33}$$

$$\frac{\mathrm{d}H_k}{\mathrm{d}t} = a_k \frac{\mathrm{d}E_{\mathrm{cum}}}{\mathrm{d}t} - \frac{H_k}{\zeta_k}, \qquad k = 1, 2, 3. \tag{S34}$$

Investment uses aggregate net output y_{n}, not the mean of the node incomes. Eqs. (S34) are the state-space form of the impulse response Eq. (S6): together with $\Delta T = R_\infty E_{\mathrm{cum}} + \sum_k H_k$ they reproduce the warming of an arbitrary emission history. Emissions are negative when $\mu > 1$.

S5.11 Economic collapse

If $y_{\mathrm{g}} \le \varepsilon$ or $y_{\mathrm{d}} \le \varepsilon$, the step is treated as an economic collapse: abatement, taxes, emissions and all per-node incomes and damages are set to zero, $\mathrm{d}K/\mathrm{d}t = -\delta K$, and the mean utility is assigned the sentinel value $\bar{u} = U = -10^{30}$ (not multiplied by L). The climate tendencies and the carried ratios are then computed as above. This branch is not reached in any of the reference runs.

S6 Time integration

The model is integrated on the grid $t_n = t_0 + n\,\Delta t$, $n = 0, \ldots, N-1$, which for 2020–2420 with $\Delta t = 1$ yr has $N = 401$ points. The initial state is $K = K_0$ and the 2020 climate state of Section S3.3, with $q_i = Q = 1$. At each grid point the tendencies are evaluated from the state at t_n, all diagnostics are recorded, and the state is advanced with the forward Euler method:

$$K_{n+1} = K_n + \Delta t\, \dot{K}_n, \quad E_{\mathrm{cum},n+1} = \max\{0,\, E_{\mathrm{cum},n} + \Delta t\, \dot{E}_{\mathrm{cum},n}\}, \quad H_{k,n+1} = H_{k,n} + \Delta t\, \dot{H}_{k,n}. \tag{S35}$$

The recorded state at t_n is the state before that step's update; the update computed at the final grid point is not used. Cumulative emissions are kept non-negative, so net-negative emissions cannot cool the climate below that of zero cumulative emissions. No other state is constrained.

S7 Derived quantities

Aggregates. Totals are per-capita values times population: $Y_{\mathrm{g}} = y_{\mathrm{g}} L$, $Y_{\mathrm{d}} = y_{\mathrm{d}} L$, $Y_{\mathrm{n}} = y_{\mathrm{n}} L$, $E = eL$. Mean consumption and savings per capita are $(1-s) y_{\mathrm{n}}$ and $s y_{\mathrm{n}}$.

Carbon price. The reported marginal abatement cost is $\theta_1 \mu^{\theta_2 - 1}$, which equals $P = 10^f$ except where μ was capped or set to zero. In the figures, the optimal carbon price in a year is this value at that grid point (the grid is annual, so no interpolation is needed).

Consumption discount rate. The model reports the consumption discount rate given by the marginal-utility-weighted Ramsey rule. With $c_{j,n} = \max\{(1-s_n)\, y_{\mathrm{n},j}(t_n),\ \varepsilon\}$ and $M_n = \sum_j w_j\, c_{j,n}^{-\eta}$,

$$r_n = e^{\rho \Delta t}\, \frac{M_n}{M_{n+1}} - 1, \qquad r_{N-1} = r_{N-2}. \tag{S36}$$

This is the discrete form of $r = \rho - \frac{\mathrm{d}}{\mathrm{d}t} \ln \sum_j w_j\, u'(c_j)$, i.e. ρ plus η times the growth rate of consumption averaged with weights proportional to marginal utility. For a static distribution it reduces to the representative-agent rule $\rho + \eta g$.

S8 Objective

The objective is the discounted integral of population-weighted mean utility,

$$J = \int_{t_0}^{t_{\mathrm{end}}} e^{-\rho (t - t_0)}\, U(t)\, \mathrm{d}t \approx \sum_{n=0}^{N-2} \frac{\Delta t}{2} \left[e^{-\rho (t_n - t_0)} U_n + e^{-\rho (t_{n+1} - t_0)} U_{n+1} \right], \tag{S37}$$

evaluated with the trapezoidal rule on the integration grid. For the manuscript configuration $J \approx -6.5 \times 10^9$.

S9 Control trajectories

Each control, $f(t)$ and $s(t)$, is represented by its values at n control times and interpolated between them.

S9.1 Control times

Control times are concentrated early in the century, where the optimal controls change fastest. In relative time, with $t_0 = 0$ and $t_{\mathrm{end}} = 400$:

1. One wind-down point is used if $n \le 10$ and two otherwise (n_{end}). The wind-down period begins at $t_{\mathrm{join}} = t_{\mathrm{end}} - 4/\delta$ (360, i.e. 2380, for $\delta = 0.1$ yr^{-1}), four e-folding times of capital depreciation before the end of the run.

2. The remaining $n_1 = n - n_{\mathrm{end}} \ge 3$ points span $[t_0, t_{\mathrm{join}}]$ with $m = n_1 - 1$ geometrically growing intervals. The first interval is Δt, and the growth factor $1 + r_0$ solves

$$\sum_{i=0}^{m-1} \Delta t\, (1 + r_0)^i = t_{\mathrm{join}} - t_0, \tag{S38}$$

   found with Brent's method on $r_0 \in (-0.5, 100)$. If the bracket contains no root, $r_0 = (t_{\mathrm{join}} - t_0)/(m \Delta t) - 1$ is used instead.

3. The points are $t_0^{\mathrm{c}} = t_0$ and $t_{i+1}^{\mathrm{c}} = t_i^{\mathrm{c}} + \Delta t (1 + r_0)^i$, accumulated sequentially, with the last point then set exactly to t_{join}.

4. The wind-down points are t_{end} if $n_{\text{end}} = 1$, or $t_{\text{end}} - 1/\delta$ and t_{end} if $n_{\text{end}} = 2$.

Table S4 lists the resulting grids. For $n = 4$, the fallback r_0 is used; it does not matter, because the second point is $t_0 + \Delta t$ for any r_0 and the third is set to t_{join}.

Table S4: Control times (calendar years) of the five refinement iterations.

n	r_0	Control times
4	179 (fallback)	2020, 2021, 2380, 2420
6	5.744311	2020, 2021, 2027.744, 2073.230, 2380, 2420
9	1.443996	2020, 2021, 2023.444, 2029.417, 2044.015, 2079.693, 2166.891, 2380, 2420
15	0.555777	2020, 2021, 2022.556, 2024.976, 2028.742, 2034.600, 2043.715, 2057.895, 2079.957, 2114.279, 2167.677, 2250.753, 2380, 2410, 2420
24	0.236343	2020, 2021, 2022.236, 2023.765, 2025.655, 2027.991, 2030.880, 2034.451, 2038.867, 2044.326, 2051.075, 2059.419, 2069.736, 2082.490, 2098.259, 2117.755, 2141.859, 2171.660, 2208.504, 2254.055, 2310.372, 2380, 2410, 2420

S9.2 Interpolation

Between control times, each control follows the shape-preserving piecewise cubic Hermite interpolant (PCHIP) of Fritsch and Butland [9, 10], as implemented in SciPy 1.15. For control points (t_k^{c}, v_k) with $h_k = t_{k+1}^{\text{c}} - t_k^{\text{c}}$ and secant slopes $m_k = (v_{k+1} - v_k)/h_k$, the derivative at an interior point is zero if m_{k-1} and m_k differ in sign or either is zero, and otherwise the weighted harmonic mean

$$d_k = \frac{w_1 + w_2}{w_1/m_{k-1} + w_2/m_k}, \qquad w_1 = 2h_k + h_{k-1}, \quad w_2 = h_k + 2h_{k-1}. \tag{S39}$$

At the first point,

$$d_0 = \frac{(2h_0 + h_1)\, m_0 - h_0\, m_1}{h_0 + h_1}, \tag{S40}$$

set to zero if its sign differs from that of m_0, and to $3m_0$ if m_0 and m_1 differ in sign and $|d_0| > 3|m_0|$; the last point is treated symmetrically, and with only two points both derivatives equal the secant slope. On $[t_k^{\text{c}}, t_{k+1}^{\text{c}}]$ the interpolant is the cubic Hermite polynomial with values v_k, v_{k+1} and derivatives d_k, d_{k+1}. Beyond the last control time the control is held at its last value. No bounds are imposed on the interpolated trajectory; the bounds of Section S10 apply only to the values at the control times.

S10 Optimization

S10.1 Refinement schedule

The optimization proceeds through $K_{\text{it}} = 5$ iterations with progressively finer control grids and larger evaluation budgets. For an initial value a and a final value b, iteration $k = 1, \ldots, K_{\text{it}}$ uses

$$\text{round}\left(1 + (a-1)\,\left[\frac{b-1}{a-1}\right]^{(k-1)/(K_{\text{it}}-1)}\right), \tag{S41}$$

with ties rounded to even. With $a = 4$ and $b = 24$ control points, the iterations use 4, 6, 9, 15 and 24 points for each control; with budgets from 2 000 to 100 000 objective evaluations, they allow 2 000, 5 317, 14 140, 37 602 and 100 000 evaluations.

S10.2 Iterations

In the first iteration, every control value is set to its initial guess ($f = 2.5$, i.e. \$316 $\mathrm{tCO_2^{-1}}$; $s = 0.256$), except at the final control time, 2420, where both are set to zero. In each later iteration, the previous optimum is interpolated (Section S9.2) onto the new control times. The optimization variables are the values of f and then of s at their control times, with bounds $f \in [-1, 4]$ (a carbon price between \$0.10 and \$10 000 $\mathrm{tCO_2^{-1}}$) and $s \in [0, 1]$; the starting point is clipped to the bounds.

Each iteration maximizes J with the Subplex algorithm [7] as implemented in the NLopt library [6] (algorithm `LN_SBPLX`). The only stopping criteria are an absolute tolerance of 10^{-8} on every variable and the iteration's evaluation budget; relative and absolute tolerances on the objective are disabled, and all other settings, including the initial step size, are NLopt's defaults. The result of the final iteration is the reported optimum. In the nine reference runs the final iteration always terminated on the variable tolerance.

S10.3 Objective presented to the optimizer

The optimizer maximizes $10^{-13}J$. If the model cannot be evaluated for a trial trajectory—because the required tax revenue cannot be raised, the Gini coefficient exceeds G_b, or the tax root solve fails—J is set to -10^{30}, which the optimizer sees as -10^{17}. If J is not finite, the optimizer receives -10^{30} directly.

S11 Simulation experiments

All 14 801 manuscript runs share the template of Section S3 and differ only in η, ρ, τ and β, with the exceptions noted. Unless stated otherwise, $\eta = 1.5$ and $\rho = 0.025$ $\mathrm{yr^{-1}}$, and the three tax cases are $\tau \in \{0, 0.5, 0.998\}$. Table S5 summarizes the families of runs. The configuration file of every run is archived with the code, so every simulation behind the manuscript can be re-run.

Table S5: Families of simulation runs.

Family	Runs	Design
Reference cases	9	$\tau \in \{0, 0.5, 0.998\} \times \beta \in \{0, 0.5, 1\}$.
Contour grid	1 681	$\tau \in \{k/40 : k = 0, \ldots, 39\} \cup \{0.998\} \times \beta \in \{k/40 : k = 0, \ldots, 40\}$.
Monte Carlo	$2 \times 1\,500$	For each $\beta \in \{0, 0.5\}$, a separate Latin hypercube sample [11] of 500 (η, ρ) pairs with $\eta \in [1, 2]$ and $\rho \in [0, 0.05]$ $\mathrm{yr^{-1}}$ (rounded to six decimals), each run for the three values of τ.
Single-parameter η	$2 \times 1\,500$	500 values of $\eta \in [1, 2]$, one in each stratum of width 0.002, with $\rho = 0.025$; $\beta \in \{0, 0.5\}$, three values of τ. The same η values are used for both β.
Single-parameter ρ	1 666 + 1 500	500 values of $\rho \in [0, 0.05]$ $\mathrm{yr^{-1}}$, one in each stratum of width 10^{-4}, with $\eta = 1.5$; $\beta \in \{0, 0.5\}$, three values of τ. For $\beta = 0$, 166 of the $\tau = 0$ cells were run twice.

Family	Runs	Design
Low-ρ extension	340	$\tau = 0$, $\beta = 0$; 340 equally spaced values of ρ from -0.029 to 0.004 yr^{-1}, extending the single-parameter ρ family to low consumption discount rates.
DICE comparison	31	$G(t) \equiv 0$, $\tau = 0$, $\beta = 0$; 20 equally spaced values of ρ from -0.015579 to 0.024421 yr^{-1}, $\rho \in \{0.03, 0.03616\}$, and nine values of ρ chosen to give $r_\mu(2030) \approx 0.8$, 1.0 (twice), 1.2, 2, 3, 4, 4.7 and 5% yr^{-1}.
Infill	2–484 per family	Additional (η, ρ) samples for Fig. 5, targeted at bands of r_μ, with $\eta \in [1, 2]$ and ρ down to -0.041 yr^{-1}; $\tau \in \{0.5, 0.998\}$ or $\tau = 0.5$ only. Some were re-run from warm starts to move runs out of a local optimum; two runs used $s \in [0.2, 0.4]$.

S12 Estimation of exogenous inputs

The estimation procedures below were run once, offline; their results were copied into the model configuration (Table S6).

Table S6: Provenance of the fitted inputs.

Model input	Source	Used as fitted?
Lorenz exponents p_k, weights γ_k	S12.1, S12.2	weights in full; exponents rounded to 6 decimals
Gini trajectory $G(t)$	S12.3	rounded to 5 decimals
$L(t)$, $\sigma(t)$	S12.4	rounded to 4–6 significant figures
$A(t)$	S12.4 (closed form)	—
θ_1, ξ, E_{land}, scalar parameters	S12.5	—

S12.1 Global income distribution

The global income distribution was constructed from the World Bank Poverty and Inequality Platform [12] (downloaded 28 December 2025: 2 556 survey rows for 172 countries, 1963–2024). Rows with complete decile shares, population and per-capita GDP were retained, and for each country the most recent survey year (1992–2024) was used. Each country–decile was treated as a group containing one tenth of the country's population with income per person

$$y_{cd} = 10\, s_{cd}\, \mathrm{GDP}_c, \tag{S42}$$

where s_{cd} is the decile's share and GDP_c the per-capita GDP reported by the platform. The 1 720 groups were sorted by income; cumulative population and income fractions, with the origin prepended, define the empirical Lorenz curve. Its Gini coefficient, $G = 1 - 2\int_0^1 \mathcal{L}\,\mathrm{d}p$ by the trapezoidal rule, is 0.680860, for a population of 7.719×10^9 and mean income \$11 627.

S12.2 Lorenz curve fit

The base Lorenz curve, Eq. (S9), was fitted to the 1 721 points (x_j, y_j) of the empirical curve (including the origin) by unweighted least squares, under the constraints $\gamma_k \geq 0$ and $\sum_k \gamma_k = 1$:

$$\min_{p} \; \min_{\gamma} \; \sum_{j} \left(y_j - \sum_{k=0}^{3} \gamma_k x_j^{p_k} \right)^2 . \tag{S43}$$

For fixed exponents, the weights were found by sequential quadratic programming (SLSQP [13]; starting point $\gamma_k = 0.25$, function tolerance 10^{-15}, at most 2 000 iterations, finite-difference gradients with step 10^{-10}). The exponents were optimized over the inner optimum with L-BFGS-B [14], starting from $p = (1.1, 7.4, 13.7, 20)$ (equally spaced), with bounds $1.01 \leq p_k \leq 1000$ and tolerances 10^{-12} on the function and the projected gradient. The fit converged in 49 iterations to a residual sum of squares of 1.811×10^{-3} ($R^2 = 0.99997863$, maximum absolute error 0.0048), giving $p = (1.50004,\ 4.36744,\ 14.0720,\ 135.060)$ and $\gamma = (0.159871,\ 0.377619,\ 0.367125,\ 0.0953854)$. In the model the exponents are rounded to six decimals and γ_0 is set to $1 - \gamma_1 - \gamma_2 - \gamma_3$. The optimum is flat: other starting points change the exponents by up to 1.1×10^{-4} (relative) at the same residual.

S12.3 Global Gini trajectory

The trajectory $G(t)$ was derived from national projections of income decile shares consistent with the Shared Socioeconomic Pathways [15], together with SSP projections of population and GDP per capita. For each SSP and for each of the 18 five-yearly years from 2015 to 2100, the 165 countries with complete data were split into deciles with population $P_c/10$ and income per person $10\, s_{cd}(t)\, \mathrm{GDP}_c(t)$, sorted by income, and the global Gini coefficient was computed with the trapezoidal rule as above (here without prepending the origin). For SSP2 this gives $G = 0.5962$ (2015), 0.5651 (2020), 0.5244 (2050) and 0.4647 (2100).

The model's $G(t)$ is an unweighted least-squares fit of Eq. (S1) to all 18 SSP2 values, with $t_r = t - 2020$ (so 2015 enters as $t_r = -5$), which gives $a = 0.19517$, $b = -0.01058$ yr^{-1} and $c = 0.37796$.

S12.4 Productivity, population and carbon intensity

$A(t)$, $L(t)$ and $\sigma(t)$ follow DICE-2023 [1], whose recurrences run on five-year steps $k = 1, 2, \ldots$, with $t_r = 5(k-1)$. Each recurrence corresponds to a Gompertz function (Eq. S3), exactly or approximately.

Population. DICE's $L_{k+1} = L_k\, (L_\infty / L_k)^{0.145}$ is exactly Gompertz, with $c = \ln(1 - 0.145)/5 = -0.031331$ yr^{-1}, $L_0 = 7752.9$ million and $L_\infty = 10\,825$ million.

Carbon intensity. DICE's $\sigma_{k+1} = \sigma_k \exp(5 g_k)$, with $g_k = -0.015 \cdot 0.96^{k-1}$, is exactly Gompertz with $c = \ln(0.96)/5 = -0.0081644$ yr^{-1} and $\sigma_\infty = \sigma_0\, e^{5(-0.015)/(1-0.96)} = \sigma_0\, e^{-1.875}$, until 2155, when DICE's floor $g_k \geq -0.005$ starts to bind. The model uses the Gompertz form throughout; it departs from DICE by about 5% in 2210 and 25% in 2280.

Productivity. DICE's $A_{k+1} = A_k/(1 - g_{A,k})$ with $g_{A,k} = 0.066\, e^{-5 \cdot 0.0015(k-1)}$ is Gompertz in the limit of small growth rates. The model uses that continuous approximation, $c = -0.0015$ yr^{-1} and

$$A_\infty = A_0 \exp\left(\frac{0.066}{5 \times 0.0015}\right) = A_0\, e^{8.8}, \tag{S44}$$

with A_0 calibrated to DICE's 2020 gross output, $A_0 = Q_0/(K_0^\alpha L_0^{1-\alpha}) = 739.619$ for $Q_0 = 135.7$ trillion dollars, $K_0 = 295$ trillion dollars and $L_0 = 7.7529 \times 10^9$. Relative to the DICE series, the model's $A(t)$ is lower by 1.45% in 2050, 3.57% in 2100 and 11.4% in 2420.

Least-squares fits of the Gompertz form to the DICE series (initial value fixed; A and σ fitted in logarithms, σ only to 2155) recover these values: $c = -0.0313308$ for L and $c = -0.0081644$, $\sigma_\infty = 4.4680 \times 10^{-5}$ for σ.

S12.5 Other inputs

- *Land-use emissions*: $5.9 \times 10^9\, e^{-0.02107\, t_r}$ tCO$_2$ yr^{-1}, DICE-2023's initial land-use emissions declining by 10% per five-year period ($\ln(0.9)/5 = -0.02107$).
- *Emission ratio*: $\xi = 1.4\, e^{-0.002085\, t_r}$, which falls to 1.185 in 2100. (DICE-2023's ratio declines linearly from 1.40 in 2020 to 1.21 in 2100 and is constant thereafter.)
- *Backstop price*: a two-term exponential fit to the DICE-2023 backstop price.
- *Scalar parameters*: α, δ, ψ_2, θ_2 and K_0 are DICE-2023 values.
- *Temperature response*: the median model of Ricke and Caldeira [3].

S13 Construction of the figures

Throughout, the optimal carbon price in 2030 is the marginal abatement cost (Section S7) at $t = 2030$, and the warming in 2100 is ΔT at $t = 2100$. Runs are matched to target values of τ or β within ± 0.01.

S13.1 Fig. 1: distribution of damage and tax burdens

Fig. 1 is analytic. Income at rank F is $y(F) = \mathcal{L}'(F)$ with $G = 0.57$, evaluated on 2 000 equally spaced ranks in $[10^{-6}, 1 - 10^{-6}]$. With damage proportional to $y \cdot y^{-\beta}$, the cumulative damage share is

$$D_\beta(F) = \frac{\int_0^F y^{1-\beta}\, \mathrm{d}F'}{\int_0^1 y^{1-\beta}\, \mathrm{d}F'}, \tag{S45}$$

and the cumulative tax share is $T_\tau(F) = \int_0^F [y - c(y)]\, \mathrm{d}F' / \int_0^1 [y - c(y)]\, \mathrm{d}F'$, with c the schedule Eq. (S21) for $\eta = 1.5$ and ΔU chosen (Brent's method, tolerance 10^{-12}) so that the average tax rate is 20%. The integrals are left Riemann sums on the rank grid. Curves are shown for $\beta \in \{0, 0.5, 1\}$ and $\tau \in \{0, 0.5, 0.998\}$.

S13.2 Fig. 2: reference cases

Fig. 2 shows the carbon price and warming from 2025 to 2100 for the six reference cases with $\beta \in \{0, 0.5\}$, $\tau \in \{0, 0.5, 0.998\}$. For each τ, the band between $\beta = 0$ and $\beta = 0.5$ is shaded.

S13.3 Fig. 3: dependence on τ and β

The 2030 carbon price and 2100 warming of the 1 681 contour-grid runs form a 41×41 grid in (τ, β). For plotting, the grid values were interpolated to 400×400 points with a piecewise cubic (Clough–Tocher) interpolant on a Delaunay triangulation. Filled contours are drawn at intervals of \$10 $\mathrm{tCO_2^{-1}}$ from \$50 to \$350 $\mathrm{tCO_2^{-1}}$ (values clipped to that range, lines every \$20) and of 0.1 °C from 1.5 to 3.5 °C. The strip $0.998 < \tau \leq 1$, which was not simulated, is shaded grey. All runs are used, without a convergence filter.

S13.4 Fig. 4 and Fig. S1: sensitivity to η and ρ

Fig. 4 uses the Monte Carlo and single-parameter families. A run is counted as converged if the optimizer reported success with termination on the variable or function tolerance; other runs are discarded. Where a parameter combination was run twice, the converged run is kept, and between two converged runs the one that used more evaluations. Every (β, τ) cell of every family retains 500 runs.

For each family, β and τ, the distribution of the 2030 carbon price and of the 2100 warming is shown as a Gaussian kernel density estimate with Silverman's bandwidth [16], evaluated at 500 points over an axis range shared across rows. A dashed line marks the central case, the run closest to $(\eta, \rho) = (1.5, 0.025)$ in the metric $\sqrt{(\eta - 1.5)^2 + (\log_{10}\rho - \log_{10} 0.025)^2}$. For the Monte Carlo family with $\beta = 0$, this is $\eta = 1.509987$, $\rho = 0.024683$, with 2030 carbon prices of \$42.79, \$71.11 and \$144.96 $\mathrm{tCO_2^{-1}}$ for $\tau = 0$, 0.5 and 0.998. Fig. S1 plots the same runs as the 2030 carbon price against the 2100 warming.

S13.5 The consumption discount rate r_μ

Figs. 5 and S3 are plotted against the consumption discount rate in 2030,

$$r_\mu = \rho + \eta \sum_j \varpi_j \ln \frac{c_j(2031)}{c_j(2030)}, \qquad \varpi_j = \frac{w_j\, c_j(2030)^{-\eta}}{\sum_k w_k\, c_k(2030)^{-\eta}}, \tag{S46}$$

with $c_j = (1 - s)\, y_{\mathrm{n},j}$: ρ plus η times the marginal-utility-weighted logarithmic growth rate of consumption. For $G = 0$ it reduces to $\rho + \eta \ln[\bar{c}(2031)/\bar{c}(2030)]$.

S13.6 Fig. 5: carbon price against r_μ

Fig. 5 combines the Monte Carlo, single-parameter and infill runs with $1 \leq r_\mu \leq 6\%\ \mathrm{yr^{-1}}$ (5 254 runs), grouped by β and τ. Where the same parameters were run in more than one family, one run was kept; this is how warm-started re-runs replace runs that had stopped in a local optimum. The 2030 carbon price is shown on a logarithmic axis.

S13.7 Fig. S2: consumption growth across the distribution

For the base case ($\tau = 0$, $\beta = 0$), the consumption growth rate of node j is $g_j(t) = 100\,[c_j(t + 1)/c_j(t) - 1]$ % $\mathrm{yr^{-1}}$, with $c_j = (1 - s)\, y_{\mathrm{n},j}$. In 2030 the population-mean growth rate is $\bar{g} = \sum_j w_j g_j = 1.794\%\ \mathrm{yr^{-1}}$ and the marginal-utility-weighted rate is $g_\mu = \sum_j \varpi_j g_j = 2.142\%\ \mathrm{yr^{-1}}$, with ϖ_j as in Eq. (S46); the poorest node grows at 2.779% and the richest at $1.065\%\ \mathrm{yr^{-1}}$. The contour panel interpolates $g_j(t)$ for 2030–2100 (71 years × 32 nodes) to 400×400 points by cubic interpolation, smooths it with a Gaussian filter (standard deviations of about 2.1 yr and 2% of rank), and draws contours from 1.0 to $3.0\%\ \mathrm{yr^{-1}}$.

S13.8 Fig. S3: comparison with DICE-2023

Fig. S3 shows the 2030 carbon price against r_μ for (i) the 31 runs with $G \equiv 0$ (DICE comparison family), interpolated with PCHIP through the 30 runs with $r_\mu \leq 5.1\%$ yr^{-1}; (ii) the 840 runs with the empirical Gini trajectory, $\tau = 0$ and $\beta = 0$ (single-parameter ρ and low-ρ families; where a run was duplicated, the one with the higher objective was kept), interpolated with PCHIP through the 567 runs with $0.5 \leq r_\mu \leq 5.5\%$ yr^{-1} and evaluated at $r_\mu = 1, \ldots, 5\%$; and (iii) the DICE-2023 carbon prices in 2030 at constant discount rates of 1–5% (\$595, 224, 114, 66 and 43 tCO$_2^{-1}$, from the DICE-2023 workbook [1]). Table S7 gives the plotted values.

Table S7: Optimal 2030 carbon price (\$ tCO$_2^{-1}$) against the consumption discount rate r_μ, as plotted in Fig. S3.

r_μ (% yr^{-1})	1	2	3	4	5
This model, $G = 0$	396.40	213.99	112.12	62.33	38.43
This model, empirical Gini	650.94	380.95	204.74	107.17	59.65
DICE-2023	595	224	114	66	43

S14 Verification

The repository contains a regression suite that compares the model with a baseline computed by the code used for the manuscript: forward replays of the nine reference runs with their published controls (relative tolerance 10^{-12} for $\tau = 0$ and 10^{-9} for $\tau > 0$, set by the tax root solve), the published results files themselves, a DICE-comparison run, six cases exercising the options not used in the manuscript, three short optimizations, and the individual numerical kernels.

Table S8 gives reference values from the replay of the base case ($\tau = 0$, $\beta = 0$; $J = -6.487938701598822 \times 10^9$). For $\tau = 0.5$, $\beta = 0.5$, $J = -6.508855233019977 \times 10^9$; for $\tau = 0.998$, $\beta = 1$, $J = -6.510344441554529 \times 10^9$.

Table S8: Reference values from the replay of the base case ($\tau = 0$, $\beta = 0$).

Quantity	2020	2030	2100
f	1.543376572092825	1.629652538702295	2.395393167065027
s	0.4786271018324223	0.2296397516562726	0.2300201664610637
K (\$)	2.95×10^{14}	$3.402759806869976 \times 10^{14}$	$1.343287212994556 \times 10^{15}$
ΔT (°C)	1.16966	1.470360532030760	3.107513349030306
y_{n} (\$)	$1.741645953981179 \times 10^4$	$2.010579385432332 \times 10^4$	$6.519107748253394 \times 10^4$
μ	$3.996318342682483 \times 10^{-2}$	$1.280010855705555 \times 10^{-1}$	$6.313729676087462 \times 10^{-1}$
Ω	$4.743218355585199 \times 10^{-3}$	$7.495515646431158 \times 10^{-3}$	$3.347956815633018 \times 10^{-2}$
U	$-2.360533615801890 \times 10^8$	$-1.915743595569843 \times 10^8$	$-1.180399699134512 \times 10^8$

S15 Numerical properties of the method

The following properties are part of the method as used for the published results.

1. *Lagged damage distribution.* Each step applies the previous step's distribution of damage across incomes, scaled to the current aggregate damage (Section S5.2). The first step uses uniform damage.

2. *Mean pre-tax income.* The node incomes $y_{\mathrm{pre},i}$ average to y_{d} only approximately (Section S5); the closed form of the flat tax assumes that they do.

3. *Quadrature error.* With 32 nodes, the mean of the empirical Lorenz density is in error by 3–7 $\times 10^{-7}$ (relative).

4. *Safeguards.* Damage fractions are confined to $[0, 1-\varepsilon]$, abatement expenditure to at most $(1-\varepsilon)y_{\mathrm{d}}$, consumption to at least ε in the utility, and the flat tax rate to at most $1-\varepsilon$, with $\varepsilon = 10^{-12}$ in absolute terms for quantities of different scale (fractions and dollars per person). The flat-tax case is selected by $|\eta_\tau - 1| < \varepsilon$, whereas logarithmic utility is used only for $\eta = 1$ exactly.

5. *Infeasible trajectories* receive the values -10^{17} or -10^{30} at the optimizer interface (Section S10.3).

6. *State constraints.* $E_{\mathrm{cum}} \geq 0$ is the only constraint on the state; μ is effectively unbounded ($\mu_{\max} = 10^{20}$).

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